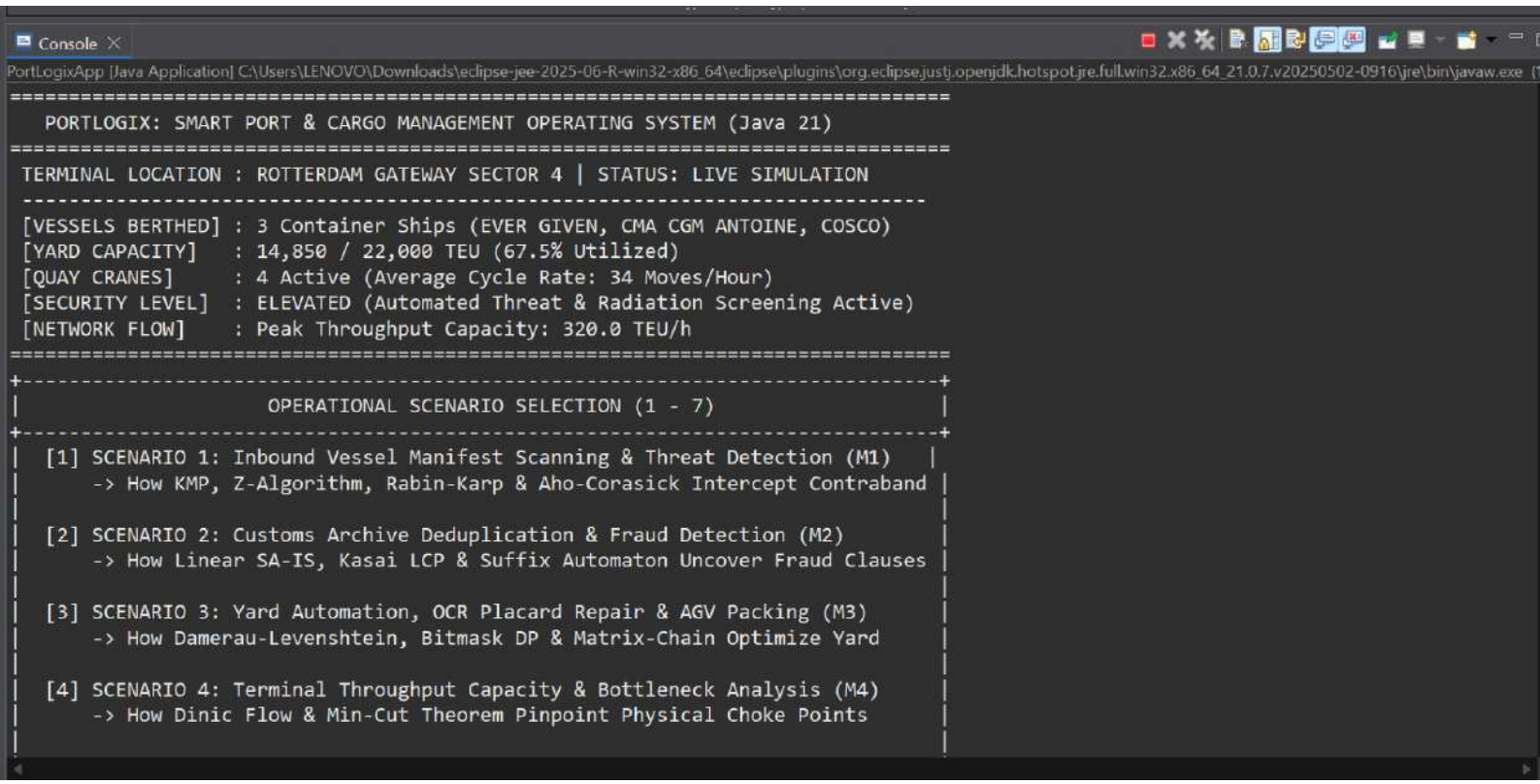

Console
PortLogixApp [Java Application] C:\Users\LENOVO\Downloads\eclipse-jee-2025-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.7.v20250502-0916\jre\bin\javaw.exe
PORTLOGIX: SMART PORT & CARGO MANAGEMENT OPERATING SYSTEM (Java 21)
TERMINAL LOCATION : ROTTERDAM GATEWAY SECTOR 4 | STATUS: LIVE SIMULATION
[VESSELS BERTHED] : 3 Container Ships (EVER GIVEN, CMA CGM ANTOINE, COSCO)
[YARD CAPACITY] : 14,850 / 22,000 TEU (67.5% Utilized)
[QUAY CRANES] : 4 Active (Average Cycle Rate: 34 Moves/Hour)
[SECURITY LEVEL] : ELEVATED (Automated Threat & Radiation Screening Active)
[NETWORK FLOW] : Peak Throughput Capacity: 320.0 TEU/h
OPERATIONAL SCENARIO SELECTION (1 - 7)
[1] SCENARIO 1: Inbound Vessel Manifest Scanning & Threat Detection (M1)
-> How KMP, Z-Algorithm, Rabin-Karp & Aho-Corasick Intercept Contraband
[2] SCENARIO 2: Customs Archive Deduplication & Fraud Detection (M2)
-> How Linear SA-IS, Kasai LCP & Suffix Automaton Uncover Fraud Clauses
[3] SCENARIO 3: Yard Automation, OCR Placard Repair & AGV Packing (M3)
-> How Damerau-Levenshtein, Bitmask DP & Matrix-Chain Optimize Yard
[4] SCENARIO 4: Terminal Throughput Capacity & Bottleneck Analysis (M4)
-> How Dinic Flow & Min-Cut Theorem Pinpoint Physical Choke Points

```
|  [2] SCENARIO 2: Customs Archive Deduplication & Fraud Detection (M2)          |
|      -> How Linear SA-IS, Kasai LCP & Suffix Automaton Uncover Fraud Clauses   |
|                                                                                |
|  [3] SCENARIO 3: Yard Automation, OCR Placard Repair & AGV Packing (M3)        |
|      -> How Damerau-Levenshtein, Bitmask DP & Matrix-Chain Optimize Yard       |
|                                                                                |
|  [4] SCENARIO 4: Terminal Throughput Capacity & Bottleneck Analysis (M4)       |
|      -> How Dinic Flow & Min-Cut Theorem Pinpoint Physical Choke Points        |
|                                                                                |
|  [5] SCENARIO 5: Hazardous Safety Compliance & 2-Approx Scanner Layout (M5)    |
|      -> How 3-SAT DPLL & Karp Reductions Guide Surveillance Placement          |
|                                                                                |
|  [6] SCENARIO 6: Big Data Telemetry, Parallel Scans & Crane Scheduling (M6)    |
|      -> How Blelloch Scan, ForkJoin Reduce & Brent's Theorem Power Real-Time   |
|                                                                                |
|  [7] SCENARIO 7: Full Smart-Port Mega-Vessel Turnaround Simulation (M1-M6)     |
|      -> Complete End-to-End Orchestration of All 6 Modules for 20,000 TEU      |
+--------------------------------------------------------------------------------+
|  [8] LAUNCH MODERN GRAPHICAL OPERATIONS DASHBOARD (Swing Desktop GUI)          |
|  [9] RUN COMPREHENSIVE ALGORITHM INTEGRITY TESTS (45 Verification Specs)       |
|  [0] SYSTEM SHUTDOWN                                                           |
+--------------------------------------------------------------------------------+
Select Command (1-7 for Scenarios, 8 for GUI, 9 for Tests, 0 to Exit): 8

>>> Launching PortLogix Graphical Operations Dashboard (Swing GUI)...
```


PortLogix: Smart Port & Cargo Management Platform (Enterprise TOS)
PORTLOGIX SMART PORT OS
• HARBOR RADAR: ACTIVE
THROUGHPUT: 320.0 TEU/H
IMDG THREAT MONITOR: ON
Enterprise Autonomous Port Control Center — Core DSA Engine (Modules M1-M6 / Scenarios 1-7)
BERTH DOCKING
3 Vessels Moored
M/V EVER GIVEN (IMO 9811000)
DISCHARGE CAPACITY
320.0 TEU / hr
Dinic Level Graph & Blocking Flow
SECURITY INTERCEPT
7 Alerts Flagged
Aho-Corasick IMDG Contraband Scanner
PARALLEL WORK-DEPTH
Max 2,500x Speedup
Brent's Theorem Task Scheduler
M3: Yard DP & Placard Auto-Repair
M5: Safety SAT & Scanner Placement
M6: Big Data Parallel Scans & Charts
7 Scenarios: Algorithm & Operational Guide
M4: Animated Terminal Flow Canvas
M1: Threat Radar & Security Scanner
M2: Customs Suffix & Fraud Indexer
MODULE 4: HIGH-THROUGHPUT FLOW NETWORK & MIN-CUT BOTTLENECK ANALYSIS
O(V²E) Dinic | O(V·E²) Edmonds-Karp
The Maritime Problem: Determine maximum container throughput (TEU/hr) from Berth to Outgate and pinpoint physical corridors restricting port expansion.
How It Works: Dinic's algorithm builds BFS level graphs and pushes DFS blocking flows in O(V²E). Max-Flow Min-Cut Theorem proves min (S,T) cut capacity exactly equals max flow, identifying saturated choke points.
Real-World Port Action: Identified 3 saturated edges from Berth 0 to Quay Cranes. Management authorizes adding 1 new Quay Gantry Crane to Berth 0 (+80 TEU/h).
Run Dinic Max Flow
Highlight Min-Cut Choke Points
Assign Cranes (Bipartite)
[Glowing Cyan = Active Flow Pipelines | Pulsing Red = Saturated Min-Cut Choke Points]
M/V EVER GIVEN
Quay Crane A
70/70
80/80
PORTLOGIX ENGINE ACTIVE
PORT SECTOR: ROTTERDAM GATEWAY 4
FLOW ALGORITHM: DINIC O(V²E)
CRYPTOGRAPHIC SEAL: 1,000,000,007 VERIFIED
SIMULATION CLOCK: LIVE 60 FPS

PortLogix: Smart Port & Cargo Management Platform (Enterprise TOS)

# PORTLOGIX SMART PORT OS

• HARBOR RADAR: ACTIVE | THROUGHPUT: 320.0 TEU/H | IMDG THREAT MONITOR: ON

Enterprise Autonomous Port Control Center — Core DSA Engine (Modules M1-M6 / Scenarios 1-7)

| BERTH DOCKING | DISCHARGE CAPACITY | SECURITY INTERCEPT | PARALLEL WORK-DEPTH |
|---|---|---|---|
| 3 Vessels Moored | 320.0 TEU / hr | 7 Alerts Flagged | Max 2,500x Speedup |
| M/V EVER GIVEN (IMO 9811000) | Dinic Level Graph & Blocking Flow | Aho-Corasick IMDG Contraband Scanner | Brent's Theorem Task Scheduler |

M3: Yard DP & Placard Auto-Repair | M5: Safety SAT & Scanner Placement | M6: Big Data Parallel Scans & Charts | 7 Scenarios: Algorithm & Operational Guide

M4: Animated Terminal Flow Canvas | **M1: Threat Radar & Security Scanner** | M2: Customs Suffix & Fraud Indexer

## MODULE 1: INBOUND MANIFEST SCANNING & AHO-CORASICK CONTRABAND RADAR

O(N + M + Z) Aho-Corasick | O(N) KMP

☁ **The Maritime Problem:** Scan gigabytes of arriving container manifest text streams in real-time to intercept explosive, toxic, and radioactive contraband.

⚙ **How It Works:** Aho-Corasick builds a Trie with BFS failure transitions & dictionary suffix links. Scans text in a single linear pass O(N + M + Z) regardless of keyword count. KMP uses pi-table to prevent backtracking.

**Real-World Port Action:** Intercepted [CLASS 1 EXPLOSIVE] and [RADIOACTIVE ISOTOPE] containers. Auto-quarantine diverted them to HAZMAT Bunker 4.

```
CARGO_MANIFEST: BL-2026-ROTTERDAM-00842 | VESSEL: EVER_GIVEN_IMO9811000
CONTAINER_1: MSKU9012347 | INDUSTRIAL SPARE PARTS AND HYDRAULIC DRILLING ACCUMULATORS.
CONTAINER_2: CMAU8765432 | ALERT: SUSPECTED CONTRABAND AND FLAMMABLE_LIQUID SOLVENTS (UN_1993).
CONTAINER_3: HLCU1122334 | CRITICAL WARNING: DETONATOR PACKAGING AND EXPLOSIVE RESIDUES REPORTED.
CONTAINER_4: MAEU7788990 | RADIOACTIVE ISOTOPE SHIELDING LEAK FLAGGED BY QUAYSIDE SENSOR.
CONTAINER_5: MSKU9012347 | REPEATED LOT VERIFICATION BILL OF LADING.
```




Search Container / SKU: MSKU9012347 | Algorithm: Aho-Corasick Multi-Pattern Security Radar | Run Threat & String Scan

| Alert Pattern | Severity | Byte Offset | Intercepted Manifest Snippet |
|---|---|---|---|
| CONTRABAND | CRITICAL | @ Byte 203 | ERT: SUSPECTED CONTRABAND AND FLAMMABLE |

PORTLOGIX ENGINE ACTIVE | PORT SECTOR: ROTTERDAM GATEWAY 4 | FLOW ALGORITHM: DINIC $O(V^2E)$ | CRYPTOGRAPHIC SEAL: 1,000,000,007 VERIFIED | SIMULATION CLOCK: LIVE 60 FPS

PortLogix: Smart Port & Cargo Management Platform (Enterprise TOS)

# PORTLOGIX SMART PORT OS

● HARBOR RADAR: ACTIVE | THROUGHPUT: 320.0 TEU/H | IMDG THREAT MONITOR: ON

Enterprise Autonomous Port Control Center — Core DSA Engine (Modules M1-M6 / Scenarios 1-7)

| BERTH DOCKING | DISCHARGE CAPACITY | SECURITY INTERCEPT | PARALLEL WORK-DEPTH |
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M3: Yard DP & Placard Auto-Repair | M5: Safety SAT & Scanner Placement | M6: Big Data Parallel Scans & Charts | 7 Scenarios: Algorithm & Operational Guide

M4: Animated Terminal Flow Canvas | M1: Threat Radar & Security Scanner | M2: Customs Suffix & Fraud Indexer

## MODULE 2: CUSTOMS DECLARATION ARCHIVING & FRAUD/PLAGIARISM INDEXER

O(N) Linear SA-IS | O(N) Kasai LCP

**The Maritime Problem:** Detect duplicate declarations, tariff fraud clauses, and plagiarized bills of lading across historical customs archives.

**How It Works:** SA-IS builds full suffix array in linear O(N) time via S/L-type induced sorting. Kasai's algorithm computes LCP array in O(N) using rank invariant h >= h - 1. Suffix Automaton DAWG matches patterns in O(M).

**Real-World Port Action:** Identified identical 38-character legal description clause between two separate declarations. Triggered audit and froze tariff exemption.

```
COMMERCIAL_BILL_OF_LADING_ROTTERDAM_INDUSTRIAL_PARTS_CLEARANCE_STAMP
```

```
CUSTOMS_ENTRY_BILL_OF_LADING_ROTTERDAM_INDUSTRIAL_PARTS_INSPECTION_PASSED
```

Build SA-IS & Kasai LCP | Detect Shared Fraud Clauses

| SA Index | Suffix Starting Pos | LCP Length | Indexed Manifest Suffix Phrase |
|---|---|---|---|
| 0 | 20 | 0 | ADING_ROTTERDAM_INDUSTRIAL_PARTS_CLEARANCE_ST... |
| 1 | 8 | 1 | AL_BILL_OF_LADING_ROTTERDAM_INDUSTRIAL_PARTS_CL... |
| 2 | 44 | 3 | AL_PARTS_CLEARANCE_STAMP |

```
>> [SA-IS Linear Time Suffix Array] 68 suffixes indexed in O(N) time.
>> [Kasai's LCP Algorithm] Longest Repeated Substring: "IAL_"
>> [Suffix Automaton DAWG] 2,284 distinct cargo phrases indexed with 91 states.
```

PORTLOGIX ENGINE ACTIVE | PORT SECTOR: ROTTERDAM GATEWAY 4 | FLOW ALGORITHM: DINIC O(V²E) | CRYPTOGRAPHIC SEAL: 1,000,000,007 VERIFIED

SIMULATION CLOCK: LIVE 60 FPS